

Experiment 3: Software Design Principles

Identify Stakeholders and Perform Requirement Elicitation

Identify stakeholders and perform requirement elicitation for the selected software project using suitable techniques such as interviews, questionnaires, observation, or user personas; prepare user stories and the Requirement Gathering Report.

Aim

To identify stakeholders and perform requirement elicitation for the **Online Mutual Fund Portal** using suitable elicitation techniques, prepare user stories, and develop a Requirement Gathering Report.

Objectives

- Identify key stakeholders of the Online Mutual Fund Portal.
- Apply suitable requirement elicitation techniques.
- Gather functional and non-functional requirements.
- Prepare user personas and user stories with acceptance criteria.
- Document the Requirement Gathering Report for the proposed system.

Introduction

Requirement elicitation is the process of discovering, understanding, and documenting the needs and expectations of stakeholders. For an online mutual fund portal, requirements must cover investor services, mutual fund information, investment management, portfolio tracking, transaction management, security, reliability, and ease of use.

Interviews, questionnaires, observation, and user personas can be combined to reduce ambiguity and improve requirement completeness.

Selected Project

Project Title Online Mutual Fund Portal

Domain Financial Technology / Investment Management

The proposed system is a web-based mutual fund portal that allows users to register and log in, browse and search mutual fund schemes, view fund details, compare different funds, invest in mutual funds, manage their investment portfolio, and view transaction and investment history.

Administrators use role-based access to manage mutual fund schemes, update fund information, manage user accounts, and monitor investments and transactions.

Stakeholder Identification

Stakeholder	Role	Key Requirements / Expectations
Investor / Customer	Uses the investment portal	Secure login, easy fund search, comparison, investment and portfolio management
Administrator	Manages portal operations	Manage mutual fund schemes, users, investments and transactions
Fund Manager / Data Provider	Provides mutual fund scheme information	Accurate and updated fund information
Payment Service	Processes investment-related payments	Secure and reliable payment processing
Database System	Stores application and investment information	Data integrity, availability and secure storage

Requirement Elicitation Techniques

Technique	Purpose	Expected Outcome
Interview	Discuss investment workflows and requirements with investors and administrators	Detailed functional requirements and constraints
Questionnaire	Collect feedback from multiple investors	User expectations, preferences and usability priorities
Observation	Observe how users search, compare and manage mutual fund investments	Workflow, usability and process requirements
User Persona	Represent typical investors and administrators	User-centred requirements and priorities

Sample Requirement Gathering / Elicitation Findings

The following findings are prepared as project-aligned sample findings for the laboratory report. If actual interviews or questionnaires are conducted later, their results should replace these sample findings.

Interview Findings

- Investors want secure and simple login functionality.

- Users need an easy way to search and filter mutual fund schemes.
- Investors require detailed information about mutual fund schemes before investing.
- Users want to compare different mutual fund schemes.
- Investors need an easy way to manage their investment portfolios.
- Users want to view their transaction and investment history.
- Administrators require facilities to add, update and remove mutual fund schemes.
- Administrators need to monitor user investments and transactions.

Observation Findings

- Users prefer simple navigation and a well-organized dashboard.
- Fund search and comparison should be easily accessible.
- Important mutual fund information should be clearly displayed before investment.
- Users should be able to access portfolio and transaction history without complicated navigation.
- Investment status and portfolio information should be clearly presented.

Questionnaire Findings

- Security is a high-priority requirement for investors.
- Users prefer a centralized platform for mutual fund discovery and investment management.
- Fund comparison is an important feature for investment decisions.
- Users value easy access to investment and transaction history.
- Investors expect the portal to be responsive and available when required.

Functional Requirements

ID	Requirement	Description
FR1	User Authentication	The system shall authenticate users and administrators using secure login mechanisms.
FR2	User Registration	The system shall allow new users to create an account.

ID	Requirement	Description
FR3	Mutual Fund Browsing	Users shall be able to browse available mutual fund schemes.
FR4	Search and Filtering	Users shall be able to search and filter mutual fund schemes.
FR5	Fund Details	Users shall be able to view detailed information about selected mutual fund schemes.
FR6	Fund Comparison	Users shall be able to compare multiple mutual fund schemes.
FR7	Investment	Users shall be able to invest in selected mutual fund schemes.
FR8	Portfolio Management	Users shall be able to view and manage their investment portfolio.
FR9	Transaction History	Users shall be able to view their previous investment transactions.
FR10	Performance Tracking	Users shall be able to track their investment performance.
FR11	Fund Management	Administrators shall be able to add, update and remove mutual fund schemes.
FR12	User Management	Administrators shall be able to manage user accounts.
FR13	Investment Monitoring	Administrators shall be able to monitor investments and transactions.

Non-Functional Requirements

ID	Category	Requirement
NFR1	Security	User and investment information shall be protected through authentication and authorization.
NFR2	Privacy	Users shall only be able to access their permitted account and investment information.
NFR3	Reliability	Investment and transaction records should be processed consistently without data loss.

ID	Category	Requirement
NFR4	Availability	The web-based portal should be available to users whenever required.
NFR5	Performance	Common operations such as fund search and portfolio viewing should provide a reasonably fast response.
NFR6	Usability	The interface should provide simple navigation and clear information.
NFR7	Scalability	The architecture should support an increasing number of users and mutual fund schemes.
NFR8	Maintainability	The system should be modular so individual components can be maintained easily.
NFR9	Auditability	Important investment and administrative actions should be traceable for system management.

User Personas

Persona	Age / Profile	Goal	Pain Point
Regular Investor	35	Find, compare and invest in mutual funds easily	Difficulty comparing multiple funds
Beginner Investor	22	Understand mutual fund schemes before investing	Lack of simple and understandable information
Experienced Investor	42	Manage multiple investments and track performance	Difficulty monitoring investments across different platforms
Administrator	40	Manage schemes, users and transactions	Managing large amounts of system information

User Stories and Acceptance Criteria

ID	User Story	Acceptance Criteria
US1	As an investor, I want to register and log in securely so that only I can access my account.	Valid credentials authenticate the user; invalid credentials are rejected.

ID	User Story	Acceptance Criteria
US2	As an investor, I want to search and filter mutual fund schemes so that I can find relevant funds easily.	Search and filter operations return matching mutual fund schemes.
US3	As an investor, I want to view mutual fund details so that I can understand a scheme before investing.	Selected fund information is displayed correctly.
US4	As an investor, I want to compare mutual funds so that I can examine different schemes.	Selected funds can be compared and relevant information is displayed.
US5	As an investor, I want to invest in a mutual fund so that I can manage my investments through the portal.	Investment request is processed and an investment record is created.
US6	As an investor, I want to view my portfolio so that I can track my investments.	The portfolio displays the user's investment records correctly.
US7	As an investor, I want to view transaction history so that I can track my previous transactions.	Previous transactions are displayed with relevant details.
US8	As an investor, I want to track investment performance so that I can monitor my investments.	Investment performance information is displayed correctly.
US9	As an administrator, I want to add and update mutual fund schemes so that fund information remains updated.	Authorized administrators can add and update valid fund information.
US10	As an administrator, I want to monitor investments and transactions so that I can manage the portal effectively.	Authorized administrators can view relevant investment and transaction records.

Requirement Gathering Report

The requirement elicitation process for the **Online Mutual Fund Portal** identified user authentication, mutual fund browsing, searching, filtering, fund comparison, investment management, portfolio tracking, transaction history and investment performance tracking as the primary investor requirements.

Administrator requirements include mutual fund scheme management, user account management, fund information management and monitoring of investments and transactions.

Stakeholder expectations emphasize **security, ease of use, accessibility, reliability, performance and maintainability**. The collected requirements provide a foundation for designing the portal's user interface, backend services, database and administrative functionality.

Observation

Using multiple elicitation techniques improves requirement completeness and reduces ambiguity. For the Online Mutual Fund Portal, combining interviews, questionnaires, observation and user personas helps capture both technical requirements and user-centred needs across investors and administrators.

Result

Stakeholders were identified, suitable requirement elicitation techniques were selected, functional and non-functional requirements were documented, user personas and user stories with acceptance criteria were prepared, and the **Requirement Gathering Report for the Online Mutual Fund Portal was successfully completed**.